

M.Sc.(I.T.) Practical Assignment 2

Q1. Write a Python program to create a Vehicle class with max_speed and mileage instance attributes. Create 5 object of the class.

Q2. Create a child class Bus that will inherit all of the variables and methods of the Vehicle . Use Setter and Getter functions.

Q3. Create a Bus class that inherits from the Vehicle class.

Give the capacity argument of Bus.seating_capacity() a default value of 50. Create appropriate function and constructor. All the function to display the capacity.

Q4. Write a program to determine which class a given Bus object belongs to.

Q5. Write a Python program to demonstrate the diamond problem.

Q6. Write a Python program to demonstrate multilevel inheritance.

Q7. Write a program to demonstrate Encapsulation and Inheritance in Python. Use Bank Account as a problem statement. This must have multiple relevant classes with applied inheritance, functions like deposit, credit, create account, delete account . Apply proper OOPS concepts .

Q8. Imagine you are building a digital banking app. You have a base Account class, and two feature classes: LoyaltyAccount (adds reward points) and FeeAccount (deducts maintenance fees).

Both feature classes override a method called apply_interest(). If a premium customer account inherits from both features using multiple inheritance, Python uses MRO and the C3 Linearization algorithm to decide which apply_interest() method runs first, preventing code conflicts.

Q9. Rewrite the previous question by making Account class as abstract class.

Q10. Demonstrate Interfaces with Q9. By adding functions like interestRate().